Cluster Innovation Centre, University of Delhi, Delhi-110007

Examination	: End Semester Examination – Dec 2025
Name of the Course	: B.Tech (Information Technology and Mathematical Innovations)
Name of the Paper	: Software Engineering
Unique Paper Code	: 3122613503
Semester	: V
Duration	: 3 hours
Maximum Marks	: 90

Instruction to students:

Attempt any SIX questions out of EIGHT. All questions carry equal marks.

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1. Answer the following questions:
   a. "Distrust in a software organization can be a serious, endemic problem." Comment.
   b. Distinguish between maturity and excellence in SDLC.
   c. List any five points Dr. Kerzner gave to project maturity
   d. Why do requirements change so much? "After all, don't people know what they want?"
   e. Distinguish between white box and black box testing.

[3x5=15]

2. Consider the following code:

```
i = 0;
n=4;
while (i<n-1) do
j = i + 1;
while (j<n) do
if A[i]<A[j] then
swap(A[i], A[j]);
j- -;
end do;
i=i+1;
end do;
```

Answer the following:
   a. Draw a flowgraph for the above code.
   b. Find the cyclomatic complexity value from the flowgraph.
   c. Find all the independent paths forming the basis set.

[5x3=15]

3. Consider the following case study and answer the questions:

"ColombaKade" is a famous hardware store for supplying building construction materials in the South and dealing with both wholesale and retail businesses. It has several branches in the southern cities. Mr. Sudantha, who is the son of the founder of this organization, is the current managing director, and he wants to expand business activities using information and communication technology (ICT). Mr. Sudantha presented his idea to the other Directors of the business at one of the recent board meetings of the directors, and they agreed to allocate some money to initiate the project. However, some directors felt that this could be a waste of money and that there would be a significant benefit to the business. Since Mr. Sudantha is a busy businessman, he appointed a small team to start the project. Mr. Lasith, who is an assistant accountant, was appointed as the leader of the team, and he was given two other members from the recruits. In addition, Mr. Sudantha will also help the team in evaluating and finalizing decisions. The project team was instructed to use ICT in three directions: give publicity to the business operations since many foreign NGOs are working in the area to reconstruct tsunami-affected houses; streamline internal management of all branches to share all resources; and provide supply chain management through an e-business model.

a) Who are the stakeholders of the "ColombaKade ICT" project? (Name at least three groups.)

b) Since the local government may provide some financial support to the project, Mr. Sudantha wanted to select someone from the accounts department as the manager of the project. Describe your views on the appointment of Mr. Lasith as the project manager, considering the nature of the work involved.

[2x7.5=15]

4. Estimate the function point value and effort for the following details.

		Weighting factors		
Measurement Parameters	**Count**	**Low**	**Average**	**High**
Number of user inputs	36	3	4	6
Number of user outputs	45	4	5	7
Number of user inquiries	48	3	6	9
Number of files	9	7	10	15
Number of external interfaces	6	5	7	10

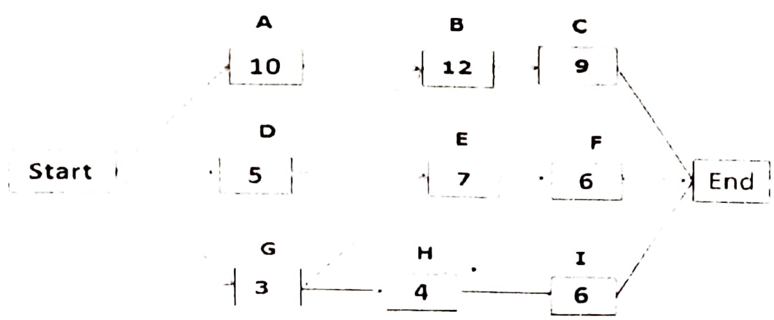
Assume the parameters are equally divided among low, average, and high complexity. Further, assume that the complexity adjustment value is 1.25.

[15]

5. Explain the Software Development Life Cycle (SDLC) using a suitable example.

[15]

6. Using the network diagram below, identify each path's total paths, critical paths, and float/slack.



[15]

7. You are given the different project activities and time allocation details in Tables 1 and 2. Estimate the expected time for each task using PERT:

Table 1. Project Activities and Time Allocation Details

Tasks	Time Allocation (person days)	Planned Start	Planned End
Requirement analysis and project planning	10	01/01/2002	01/14/2002
Setting up the environment	6	01/01/2002	01/08/2002
Software construction	80	01/15/2002	05/15/2002
Unit testing	30	05/16/2002	06/28/2002
User training	5	07/22/2002	07/26/2002
System testing	15	07/01/2002	07/19/2002
User documentation	30	01/15/2002	02/28/2002
Data migration	20	01/15/2002	02/15/2002
Conducting user acceptance test	20	08/01/2002	08/31/2002

Table 2. Optimistic, Most Likely Time, and Pessimistic Estimates for Activities

Tasks	Optimistic Time Estimates (person days)	Most Likely Time Estimates (person days)	Pessimistic Time Estimates (person days)
Requirement analysis and project planning	7	10	13
Setting up the environment	3	6	9
Software construction	48	83	100
Unit testing	20	28	33
User training	4	5	6
System testing	10	15	20
User documentation	23	28	45
Data migration	18	18	30
Conducting user acceptance test	14	21	22

[15]

8. In each SDLC phase, make memos for the team if the responsible workers have not met expectations.

[15]